ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

# INTRODUCTION TO TECHNICAL PROGRAMMING

## PAPER 1 · PRACTICE C

**MARKS: 120** **TIME: 3 hours**

**NC(V) Programming and Robotics, Level 3. Laid out like the DHET first paper (120 marks). Section A (31): multiple choice, true or false, matching C++ terms and flowchart symbols · Section B (89): programming concepts and computational thinking, Git and GitHub teamwork, an mBlock guessing game to study (with C++ and a flowchart) and a C++ battery-test program to study.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Program code in this paper is written in C++, except for the mBlock program in QUESTION 7. Code is not marked strictly on syntax.
6. Write neatly and legibly.

## SECTION A

## QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

**1.1** Which one of the following is a block-based programming environment?

A Scratch B Visual Studio Code
C GCC D Notepad

**1.2** The Raspberry Pi is best described as a …

A microcontroller only. B single-board computer.
C sensor module. D printer.

**1.3** Which C++ operator gives the remainder after a division?

A / B * C % D ++

**1.4** The Git command that copies a GitHub repository onto your own computer is …

A git push. B git commit. C git status. D git clone.

**1.5** The C++ data type used to store a single letter or symbol is …

A char. B int. C bool. D double.

**1.6** In C++, each statement ends with a …

A colon. B semicolon. C full stop. D comma.

**1.7** The flowchart symbol used for a decision is a …

A rectangle. B oval. C diamond. D parallelogram.

**1.8** The C++ expression (7 > 3 && 2 > 5) gives the value …

A true. B false. C 5. D 7.

**1.9** Breaking a big problem into smaller, easier parts is called …

A decomposition. B abstraction.
C compilation. D debugging.

**1.10** Which C++ loop always runs its body at least once?

A for B while C if D do … while

(10)

**[10]**

## QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

**2.1** A compiler translates the whole program into machine code before the program runs.

**2.2** In C++, the names Total and total refer to the same variable.

**2.3** The command git push sends your local commits to a remote repository such as GitHub.

**2.4** Pseudocode must follow the exact syntax rules of C++.

**2.5** A while loop tests its condition before every repetition.

(5)

**[5]**

## QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

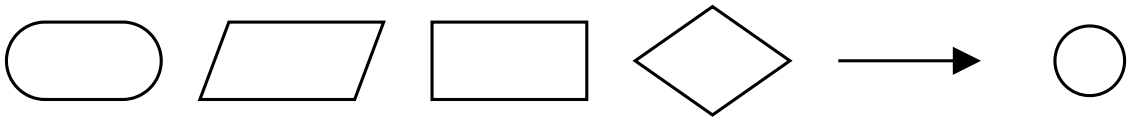
| | COLUMN A | | COLUMN B |
|---|---|---|---|
| 3.1 | Ends the output line and moves to the next line | A | switch |
| 3.2 | Data type that stores true or false | B | // |
| 3.3 | Chooses between several fixed cases | C | bool |
| 3.4 | Library needed for cin and cout | D | break |
| 3.5 | Leaves a loop immediately | E | endl |
| 3.6 | Declares a value that cannot change | F | array |
| 3.7 | Starts a one-line comment | G | #include <iostream> |
| 3.8 | Increases a variable by one | H | int |
| 3.9 | Stores several values of the same type under one name | I | const |
| 3.10 | Data type for whole numbers | J | ++ |

(10)

**[10]**

## QUESTION 4

Study the flowchart symbols below and write the name of each symbol (A–F) next to the question number (4.1–4.6).



A B C D E F

**4.1** A

**4.2** B

**4.3** C

**4.4** D

**4.5** E

**4.6** F

(6)

**[6]**

**TOTAL SECTION A: [31]**

## SECTION B

## QUESTION 5

**5.1** Answer the questions below. Write only the answer next to the question number (5.1.1–5.1.5).

5.1.1 Define the term syntax error. (2)

5.1.2 Define the term variable, and give an example of a variable declaration in C++. (4)

5.1.3 Which computer component carries out the instructions of a program? (2)

5.1.4 Name a device that stores data permanently, even when the computer is switched off. (2)

5.1.5 Define the term algorithm. (2)

**5.2** Explain what abstraction means in computational thinking. (2)

**5.3** Give the differences between high-level and low-level programming languages. (4)

**5.4** Explain the difference between a local variable and a global variable. (4)

**5.5** List FOUR steps of the program development cycle. (4)

**[26]**

## QUESTION 6

Answer the following question as fully as possible.

> Three students are building an app that keeps the scores at a college robot competition. One student writes the code that captures the scores, one writes the code that ranks the teams, and one designs the screens. They all work on the same project at the same time from their own laptops.

Explain how the students can use Git and GitHub to work together effectively on this project. (11)

**[11]**

## QUESTION 7

Study the mBlock program below and answer the questions.

```
when green flag clicked
set [secret] to (pick random 1 to 20)
set [tries] to 0
set [guess] to 0
repeat until < (guess) = (secret) >
    ask [Guess a number from 1 to 20] and wait
    set [guess] to (answer)
    change [tries] by 1
    if < (guess) > (secret) > then
        say [Too high!] for 1 seconds
    if < (guess) < (secret) > then
        say [Too low!] for 1 seconds
end repeat
say (join [Correct! Tries: ] (tries)) for 3 seconds
```

**7.1** Name the THREE variables used in the program and say what each one stores. (3)

**7.2** How does the program choose the secret number? (2)

**7.3** Which block makes the questions repeat, and when does the repeating stop? (2)

**7.4** What does the block "change tries by 1" do? (2)

**7.5** What will the sprite say if the secret number is 9 and the user enters 15? (2)

**7.6** The secret number is 12. The user enters 5, then 18, then 12. Write down everything the sprite says, in order. (3)

**7.7** Write a C++ program that plays the same game, using a do … while loop. (8)

**7.8** Explain the difference between a "repeat until" block and a "repeat 10" block, and why "repeat until" is the better choice here. (2)

**7.9** Draw a flowchart for the mBlock program above. (8)

**[32]**

## QUESTION 8

A workshop tests the batteries of its robots. A technician types in the voltage of each battery. A voltage of 7,0 V or more is good, from 6,4 V up to 7,0 V is weak, and below 6,4 V is flat. The technician types −1 when there are no more batteries.

Study the program below and answer the questions that follow.

```
 1  #include <iostream>
 2  using namespace std;
 3
 4  int main() {
 5      double volts;
 6      int good = 0, weak = 0, flat = 0;
 7      double sum = 0;
 8      int count = 0;
 9
10      cout << "Enter battery voltage (-1 to stop): ";
11      cin >> volts;
```

```
12      while (volts != -1) {
13          if (volts >= 7.0) {
14              good++;
15          } else if (volts >= 6.4) {
16              weak++;
17          } else {
18              flat++;
19          }
20          sum = sum + volts;
21          count++;
22          cout << "Enter battery voltage (-1 to stop): ";
23          cin >> volts;
24      }
25
26      if (count > 0) {
27          cout << "Batteries tested: " << count << endl;
28          cout << "Good: " << good << "  Weak: " << weak << "  Flat: " << flat << endl;
29          cout << "Average voltage: " << sum / count << " V" << endl;
30      } else {
31          cout << "No batteries tested." << endl;
32      }
33      return 0;
34  }
```

**8.1** Which loop structure is used, and what is the special value that stops it called? (2)

**8.2** Why is there one cin statement before the loop (line 11) and another at the end of the loop (line 23)? (2)

**8.3** What does the statement good++ do? (2)

**8.4** In which category will a battery of exactly 6,4 V be counted? (2)

**8.5** The technician enters 7,4 then 6,2 then 6,8 then −1. What are the values of good, weak and flat at the end? (2)

**8.6** Calculate the average voltage that is displayed for the input in QUESTION 8.5. (2)

**8.7** Why does the program check that count is greater than 0 before it calculates the average? (2)

**8.8** What is displayed if the first value entered is −1? (2)

**8.9** Give the data type of the variable volts and explain why this type is used. (2)

**8.10** Rewrite line 20 using a compound assignment operator. (2)

**[20]**

**TOTAL SECTION B: [89]**

**TOTAL: 120**

**END OF QUESTION PAPER**